

AI-Powered Personalized Course Creation and Discovery Platform with Adaptive Learning Paths

R26-SE-041

Project Proposal Report

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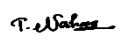
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March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT (Maximum 300 words)

The traditional "one-size-fits-all" teaching model in higher education often struggles to accommodate diverse learning paces, frequently leading to student disengagement and knowledge gaps. A significant research gap exists within current digital learning platforms; many offer only static tutoring or simple automated assessments without deeply personalizing the learning pathway based on a student's real-time understanding. Existing tools often provide generic quizzes and direct answers rather than fostering active problem-solving skills. To address this issue, my research component proposes an Adaptive Quiz and Assessment Generation Engine. When a student uploads educational materials, the system utilizes Natural Language Processing (NLP) to automatically generate diverse question types, including MCQs, fill-in-the-blanks, and short answers. The core of the system is a Knowledge Tracing Model that evaluates student interactions to infer their mastery of specific concepts. If a student answers incorrectly, the system adapts the difficulty of future questions and provides Socratic hints to guide their thinking instead of giving the direct answer. Additionally, it uses Retrieval-Augmented Generation (RAG) with a vector database to fetch targeted learning resources, such as specific YouTube tutorials and PDFs, for areas where the student is weak. The expected outcome is a highly interactive, closed-loop learning environment that demonstrably improves student comprehension, measured through pre- and post-test evaluations. Socially, this promotes equitable education by catering to individual needs and supporting localized learning. Commercially, it offers a powerful EdTech solution that can help institutions improve student retention and engagement through deep, intelligent personalization.

Keywords: Adaptive Assessment, Knowledge Tracing, NLP, RAG, EdTech.

ACKNOWLEDGEMENT

I would like to express my deepest gratitude to everyone who supported me throughout my 4th-year research project. First and foremost, I sincerely thank our Research Project supervisor, Prof. Nuwan Kodagoda, whose continuous guidance, expertise, and encouragement were invaluable in successfully advancing our work on the AI-Powered Personalized Course Creation and Discovery Platform. I am also extremely grateful for the insightful advice and constructive feedback provided by the evaluation panel throughout our project assessments. Furthermore, I would like to extend my sincere thanks to our co-supervisor, Ms. Malithi Nawarathne, for their valuable recommendations that helped us execute our work more efficiently.

I must also deeply acknowledge my fellow group members—Kojithan P.Y , Baskaran V , and Sarmitha S —for their exceptional teamwork, dedication, and support throughout this journey. Finally, I would like to give my heartfelt gratitude to my parents and friends who consistently stood by me and helped make this project a success.

TABLE OF CONTENT (Auto Generated)

Table of Contents

DECLARATION	2
ABSTRACT (Maximum 300 words)	3
ACKNOWLEDGEMENT	4
TABLE OF CONTENT (Auto Generated)	5
LIST OF TABLES	6
LIST OF FIGURES	7
LIST OF ABBREVIATIONS	8
LIST OF APPENDICES	9
INTRODUCTION.....	10
1.1 Background and Context.....	10
1.2 Literature Review.....	11
1.3 Research Gap	12
2. OBJECTIVE	13
2.1 Main Objective.....	13
2.2 Specific Objectives	13
3. METHODOLOGY	13
4. HIGH-LEVEL SYSTEM ARCHITECTURE	16
5. USER REQUIREMENTS.....	16
6. COMMERCIALIZATION PLAN	18
7. BUDGET AND JUSTIFICATION.....	19
8. WORK BREAKDOWN STRUCTURE (WBS).....	21
9. GANTT CHART.....	23
10. REFERENCES LIST.....	24
11. APPENDICES.....	25

LIST OF TABLES

TABLE 1: RESEARCH GAP COMPARISON	13
TABLE 2:BUDGET PLAN	19

LIST OF FIGURES

FIGURE 1:HIGH-LEVEL SYSTEM ARCHITECTURE	16
FIGURE 2:WBS	22
FIGURE 3:GANT CHART.....	23
FIGURE 4:DETAILED DIAGRAM	26
FIGURE 5:MOCKUP1	27
FIGURE 6:MOCKUP2.....	27
FIGURE 7:MOCKUP3	27

LIST OF ABBREVIATIONS

AI - Artificial Intelligence

AQG - Automatic Question Generation

B2B - Business-to-Business

B2C - Business-to-Consumer

CoEAI - Centre of Excellence for AI

IRB - Institutional Review Board

LKR - Sri Lankan Rupee

LLM - Large Language Model

LMS - Learning Management System

MCQ - Multiple Choice Question

NLP - Natural Language Processing

RAG - Retrieval-Augmented Generation

SaaS - Software as a Service

UI - User Interface

WBS - Work Breakdown Structure

LIST OF APPENDICES

APPENDIX A: SURVEY INSTRUMENTS.....	25
APPENDIX B: DETAILED DIAGRAMS.....	26
APPENDIX C: PRELIMINARY UI MOCKUPS.....	27
APPENDIX D: RISK ANALYSIS.....	28

INTRODUCTION

1.1 Background and Context

In higher education today, teaching and assessment methods often follow a traditional "one-size-fits-all" model. Usually, educators provide the same learning materials and the exact same static quizzes to an entire class. However, every student has a different learning pace and varying levels of understanding. When assessments do not adapt to these individual differences, students easily lose interest, develop knowledge gaps, and experience poor academic progression.

This problem primarily affects university students, especially undergraduates, as well as their educators. Students who come from non-traditional backgrounds or have different learning styles are disproportionately impacted by these rigid educational structures. Addressing this is highly important because inefficient learning processes leave graduates unprepared for the modern workforce and can widen the achievement gap, making education less equitable.

Globally and locally, there is strong evidence showing that student retention and engagement drop significantly when personalized support is missing. Research indicates that AI-driven adaptive learning platforms significantly outperform traditional instruction by tailoring pathways to individual needs and keeping students actively engaged.

While there are existing AI solutions in the market, they have notable limitations. For example, tools like Google NotebookLM simply read uploaded PDFs and provide static, one-time summaries in English. They do not actively test the student's knowledge or provide personalized practice materials. Other basic platforms might generate quizzes, but they give identical questions to all learners and immediately reveal the correct answers when a mistake is made. They lack the ability to track a student's ongoing performance or guide their critical thinking. Furthermore, most of these platforms are restricted to English, creating an accessibility barrier for local students who prefer learning in Sinhala or Tamil.

To solve these issues, my research focuses on developing an Adaptive Quiz and Assessment Generation Engine. When a student uploads their learning materials (like PDFs or notes), the system uses Natural Language Processing (NLP) to automatically generate various question types, including MCQs, fill-in-the-blanks, and short answers. Using a Knowledge Tracing Model, the system tracks the student's exact performance history to understand their strong and weak concepts. Based on this data, it automatically adjusts the difficulty of the quiz in real-time. If a student answers incorrectly, the system does not give the direct answer; instead, it provides Socratic hints to guide their thinking. Additionally, using Retrieval-Augmented Generation (RAG) and web scraping, the system will find targeted learning resources like YouTube tutorials and PDFs for the exact topics the student is struggling with.

This project directly aligns with Sustainable Development Goal 4 (Quality Education) by using technology to make inclusive, equitable, and highly personalized learning accessible to everyone. Furthermore, this research belongs to the Centre of Excellence for AI (CoEAI)

research cluster, utilizing core AI domains like machine learning and NLP to innovate within the EdTech industry.

1.2 Literature Review

The integration of Artificial Intelligence (AI) in education has initiated a paradigm shift from traditional teaching to personalized learning environments. However, current educational technologies remain highly fragmented. Students are often required to use different, disconnected tools for reading notes, taking quizzes, and finding video tutorials, which disrupts their learning flow. To address this, our proposed overall system aims to create a unified, multimodal learning platform where content creation, interactive exploration, and adaptive assessments operate synergistically.

Within this overarching ecosystem, dynamic assessment forms a critical pillar to evaluate and guide student progression continuously. Focusing specifically on educational assessment, early foundational studies [5] highlighted the effectiveness of knowledge-based web-enabled agents in intelligent tutoring systems. Building upon this, recent research has shifted towards predictive modeling. For instance, [8] investigated deep learning for knowledge tracing based on behavioral features, while [1] proposed a wheel-spinning detection model to identify exactly when students are stuck. While these studies accurately predict student performance, they often lack a fully integrated pipeline that actively translates these predictions into dynamically generated assessments. To bridge the gap in automated content creation, researchers [2], [9] demonstrated the massive potential of Natural Language Processing (NLP) for Automatic Question Generation (AQG). However, a critical evaluation reveals that applying AQG in isolation without real-time difficulty adjustment based on a student's cognitive state limits its true educational impact.

Beyond generating questions, the feedback mechanism is crucial for active learning. Research by [3] critically evaluated teaching methods and proved that employing Socratic questioning strategies significantly enhances students' critical thinking and problem-solving skills compared to simply providing correct answers. Furthermore, to provide actionable study support, recent frameworks [6] have shown how Retrieval-Augmented Generation (RAG) can drive adaptive quizzes, and [4] emphasized effective learning resource recommendations. Additionally, studies like [7] and [10] highlighted the necessity of adaptive decision support and visual feedback. Despite these advancements, most existing scholarly models tackle these elements knowledge tracing, question generation, and resource recommendation as completely separate research problems rather than a cohesive, real-time diagnostic loop.

In the commercial landscape, various products such as Google NotebookLM, Khanmigo, ChatGPT's Study Mode, and StudyX attempt to integrate these AI concepts by allowing users to upload documents to generate notes and basic quizzes. However, a critical evaluation reveals significant limitations within these services. Most platforms provide a static assessment experience; they generate a set of questions from a PDF but deliver the exact same quizzes to every student, failing to adapt to individual comprehension levels. Furthermore, when a student answers incorrectly, these systems typically reveal the correct answer immediately rather than guiding the student to think independently. Additionally, these tools operate as fragmented services and are predominantly restricted to the English language. Therefore, a critical limitation of both existing literature and commercial systems is the lack of a tightly integrated engine that dynamically traces knowledge, adjusts quiz

difficulty in real-time, and actively provides Socratic hints along with localized, web-scraped study resources

1.3 Research Gap

While Artificial Intelligence has significantly advanced educational content delivery, a major research gap remains in how these systems dynamically evaluate and respond to a student's evolving proficiency. Current approaches, including standard Learning Management Systems (LMS) and modern AI generators like Google NotebookLM, are insufficient because they provide a highly static assessment experience. Typically, they generate a fixed set of questions from a source document and deliver the exact same quiz to every student, failing to account for individual mastery levels. Furthermore, when a learner struggles or answers incorrectly, these platforms immediately spoon-feed the correct answer. This approach stifles independent problem-solving and critical thinking. Additionally, these systems operate almost exclusively in English, creating a barrier for students who rely on multilingual support (such as Sinhala and Tamil) and failing to actively fetch targeted study materials when a specific knowledge gap is detected.

To address these critical missing links, the novelty of the proposed Adaptive Quiz and Assessment Generation Engine lies in the tight integration of a continuous Knowledge Tracing Model with dynamic, multi-format question generation. Unlike existing systems where a quiz acts merely as a final summative test, this component creatively repurposes quizzes into real-time diagnostic probes. By continuously tracking the student's interaction history, the adaptive algorithm automatically adjusts the difficulty and topic focus of upcoming questions—challenging advanced learners while easing the cognitive load for those who are struggling. Creatively, if a student makes a mistake, the system does not immediately reveal the correct answer. Instead, it utilizes Natural Language Processing (NLP) to generate context-aware Socratic hints, actively guiding the student to think critically and arrive at the solution themselves.

This proposed solution is firmly grounded in an evidence-based approach to learning. Educational research strongly indicates that students achieve higher mastery when assessments are tailored to their specific cognitive levels and when they receive formative, guided feedback rather than passive answers. To provide concrete support for weak learning areas, this system leverages Retrieval-Augmented Generation (RAG) and targeted web scraping. Rather than leaving a struggling student stranded, it automatically fetches highly relevant external resources, such as specific YouTube video tutorials and supplementary PDFs, directly mapped to their identified knowledge gaps. By combining real-time difficulty adjustment, Socratic feedback, and localized multilingual support, this engine fills the critical gap between simple automated testing and genuine, personalized learning mastery.

Features / Systems	Existing Systems	My Proposed System
Multi-format question generation	Partial	✓
Dynamic difficulty adjustment (Knowledge Tracing)	X	✓
Context-aware Socratic hints	X	✓
Automated supplementary resource fetching (RAG)	X	✓

Multilingual support (Sinhala & Tamil)	X	✓
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Table 1: Research Gap Comparision

2. OBJECTIVE

2.1 Main Objective

To develop an AI-powered Adaptive Quiz and Assessment Generation Engine that utilizes Natural Language Processing (NLP) and a Knowledge Tracing Model to automatically generate customized question formats from learning materials and dynamically adjust quiz difficulty based on real-time student performance, aiming to measurably improve student comprehension and concept mastery as evaluated by comparing pre-test and post-test scores.

2.2 Specific Objectives

- Developing an NLP pipeline to automatically generate diverse quiz questions (MCQs, fill-in-the-blanks, etc.) from uploaded learning materials.
- Implementing a Knowledge Tracing Model to track and evaluate a student's real-time mastery of specific concepts.
- Adjusting quiz difficulty dynamically based on the student's ongoing performance and knowledge gaps.
- Incorporating a feedback system that provides Socratic hints and external study resources (via web scraping) for incorrect answers.
- Evaluating the engine's overall impact on learning efficacy by analyzing pre-test and post-test scores.

3. METHODOLOGY

The methodology for developing the Adaptive Quiz and Assessment Generation Engine involves a structured process to craft a functional software solution for a real-world educational problem. This involves defining the research approach, selecting the right algorithms, establishing system requirements, and outlining how the system will be tested and evaluated.

3.1 Research Approach

This project follows a **Design Science Research** approach. This method is highly suitable because the primary objective is to design, build, and evaluate an innovative IT artifact—in this case, an intelligent assessment system—that solves the practical problem of "one-size-fits-all" testing in higher education.

3.2 Technical Methods and Algorithms

The engine's intelligence relies on four main interconnected methods:

- **NLP Pipeline:** Natural Language Processing is used to automatically extract information from uploaded source content to generate diverse question types, including MCQs, fill-in-the-blanks, and short answers.
- **Knowledge Tracing Model:** This algorithm continuously tracks a student's interaction history during quizzes to infer their real-time mastery of specific concepts.
- **Adaptive Algorithm:** Using data from the knowledge tracing model, this logic dynamically adjusts the difficulty level and topic focus of upcoming questions.
- **Feedback Mechanism:** Instead of just giving direct answers, the system generates Socratic hints for incorrect answers and uses web scraping to find relevant external study resources.

3.3 Requirements

3.3.1 Functional Requirements

- The system must allow students to upload learning materials like PDFs or text.
- The system must automatically generate quiz questions based on the uploaded content.
- The system must evaluate student answers and update their knowledge mastery score in real-time.
- The system must adjust the difficulty of the next question based on the student's performance.
- The system must provide Socratic hints and web links for incorrect answers.

3.3.2 Non-functional Requirements

- **Performance:** The system must generate questions and evaluate answers with minimal delay.
- **Usability:** The interface must be easy to navigate for non-technical students.
- **Security:** The system must protect student data by only collecting necessary interaction logs.
- **Reliability:** The adaptive algorithm must function continuously without crashing during a quiz session.

3.3.3 User Requirements

- Students need a clear, distraction-free interface to focus on their quizzes.
- Users require proper on-screen guidance to understand how the adaptive testing works.
- Minimum response time is needed when fetching Socratic hints or external resources.

3.3.4 System Requirements

- **Software Requirements:**
 - *User-end:* Any standard Web Browser (Chrome, Safari, Edge).
 - *Developer-end:* Python (for core AI/ML logic), Node JS / Express (for backend API), React JS (for frontend), and MongoDB (Database).
- **Hardware Requirements:**
 - *User-end:* A standard PC, Laptop, or Mobile Phone with internet access.
 - *Developer-end:* A standard PC/Laptop for coding, with cloud computing support for training AI models.

3.4 Tools and Platforms with Justification

- **Python:** Chosen as the primary backend language due to its extensive support for Machine Learning and NLP libraries.
- **Hugging Face (Open-source LLMs):** Justified for use in the NLP pipeline to generate questions and Socratic hints without relying on expensive, proprietary black-box APIs.
- **MongoDB:** Selected as the database because its NoSQL structure is highly flexible for storing dynamically generated JSON quiz data and evolving learner profiles.
- **React JS:** Chosen for the frontend to build a highly responsive and interactive user dashboard.

3.5 Data Collection and Ethical Considerations

Because accessing real student data is challenging and sensitive, the project uses a hybrid data strategy:

- **Data Collection:** Initially, Large Language Models (like Gemini 3 Pro or Deepseek) will be used to generate synthetic course content and simulated student interactions. To calibrate the algorithms, public benchmarking datasets like "EdNet" and "ASSISTments" will be utilized. Finally, a controlled pilot study with volunteer undergraduates will be conducted to gather real-world data.
- **Ethical Considerations:** For the pilot study, strict ethical clearance will be obtained from the Institutional Review Board (IRB). The system will enforce Informed Consent, Data Minimization (collecting only necessary interaction logs), and the Right to Withdraw at any time.

3.6 Validation Strategy and Performance Metrics

The ultimate goal of the system is to improve student learning. Therefore, the primary validation strategy involves conducting a controlled study where the engine's impact on learning efficacy is evaluated by analyzing and comparing the students' pre-test and post-test scores.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

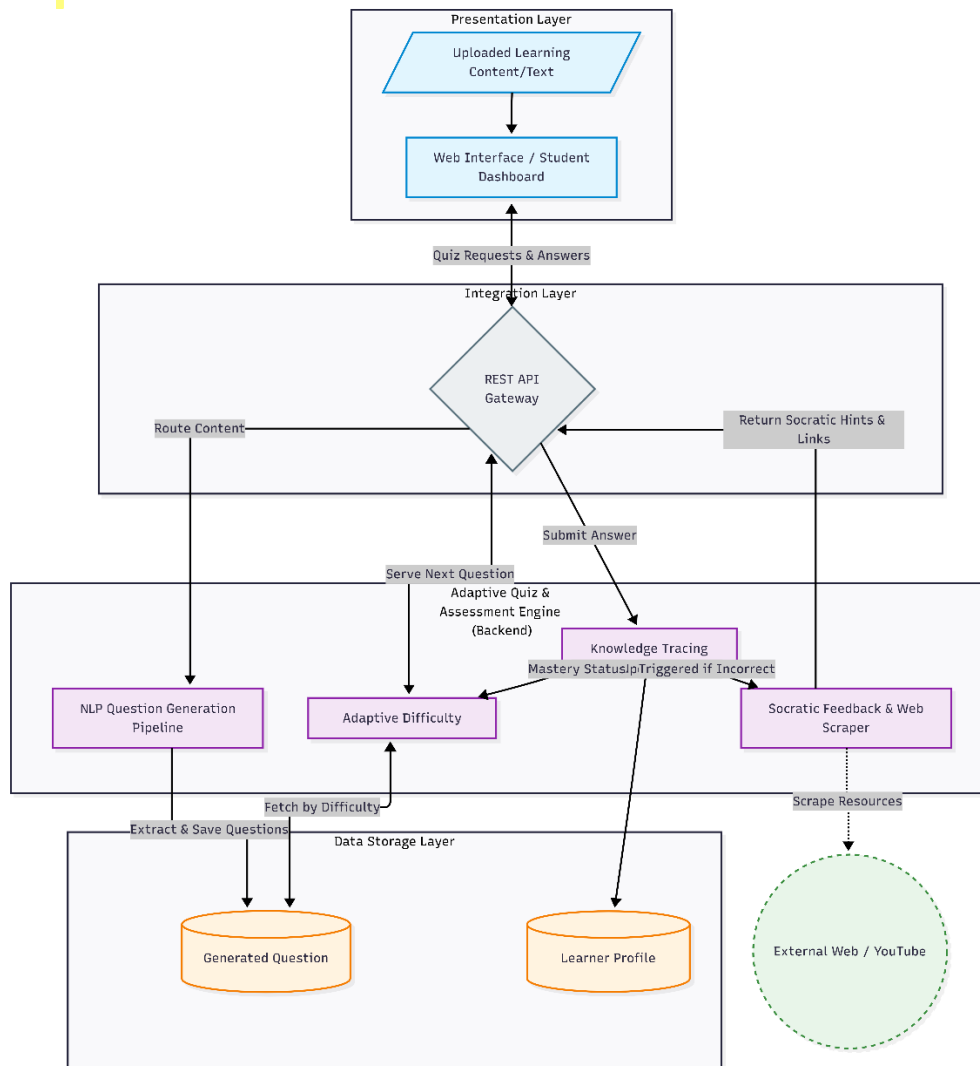


Figure 1: High-Level System Architecture

5. USER REQUIREMENTS

5.1 Stakeholder identification

The primary stakeholders for this specific component are:

- **Students (Primary Users):** University undergraduates who will upload their notes, take the adaptive quizzes, and receive personalized feedback to improve their learning.
- **Educators / Lecturers (Secondary Users):** Teachers who might upload course materials for the system to generate initial question banks for their students.

5.2 Functional requirements (numbered)

These are the core functions the system must perform for the user:

1. The system must allow users to upload learning materials (such as PDFs, documents, or plain text).
2. The system must automatically extract text and generate diverse question formats, including MCQs, fill-in-the-blanks, and short answers.
3. The system must evaluate student answers in real-time and update their individual knowledge mastery score.
4. The system must automatically adjust the difficulty level of the next quiz question based on the student's current mastery level (e.g., giving easier questions if they are struggling).
5. The system must generate and display Socratic hints rather than providing the direct correct answer when a student makes a mistake.
6. The system must automatically scrape the web and recommend targeted external study resources (like YouTube links) when a student's performance in a specific topic drops below a certain level.

5.3 Non-functional requirements (security, performance, usability, etc.)

These define how well the system should perform its functions:

- **Performance (Speed):** The system must have a fast response time. Generating questions from uploaded PDFs should be done quickly, and the transition between submitting an answer and getting the next adaptive question (or hint) must happen with minimal delay to keep the student engaged.
- **Usability:** The interface must be user-friendly, clean, and distraction-free. Furthermore, it must support multilingual interactions (English, Sinhala, and Tamil) to be accessible to local students.

- **Security and Privacy:** The system must protect student data by following data minimization principles. It should securely log only the necessary interaction data (like scores and attempt times) required for the adaptive algorithm, without exposing or misusing sensitive personal information.
- **Reliability:** The knowledge tracing model must accurately calculate mastery levels without crashing during continuous quiz sessions.

6. COMMERCIALIZATION PLAN

The commercialization of the Adaptive Quiz and Assessment Generation Engine presents a strong opportunity within the EdTech industry, specifically targeting the higher education market. The primary customer personas for this system are university undergraduates who require guided, self-paced learning, as well as educators and institutions seeking intelligent, automated assessment tools. The core value proposition of this product lies in its ability to transform traditional, static testing into a highly personalized diagnostic experience. By tightly integrating a knowledge-tracing model with multi-format question generation, the engine dynamically adjusts difficulty and provides Socratic hints along with targeted study resources instead of just giving direct answers. This deep, real-time personalization, combined with a transparent open-source architecture, provides a strong competitive advantage over existing proprietary, black-box platforms like Google NotebookLM.

To successfully bring this product to market, the revenue model will rely on a dual approach: a B2C freemium model for individual students and a B2B SaaS licensing model for universities. We will employ a penetration pricing strategy, offering highly affordable premium subscriptions to students to quickly capture market share, while charging institutions based on tiered licensing agreements. The cost estimation for operating this platform will primarily involve cloud infrastructure expenses for hosting the NLP models, database costs for securely storing learner profiles, and ongoing system maintenance. Finally, addressing intellectual property considerations is crucial; while we will strictly comply with the licensing agreements of the open-source Large Language Models we utilize, we will seek to copyright our unique custom logic, particularly the proprietary knowledge-tracing algorithm and the specific integration pipeline that drives the adaptive feedback loop.

7. BUDGET AND JUSTIFICATION

Developing the interactive infographic generator is primarily a software-based endeavor; therefore, there are no physical hardware components associated with the core implementation. To ensure the successful development and testing of the intelligent data extraction schema and visualization engine, we have planned to utilize cloud computing.

Below is the estimated budget required to complete this specific component of the research project, tailored to the constraints of an undergraduate study.

Expense Category	Estimated Amount (LKR)
Hardware Costs	0.00
Software Licenses	0.00
Cloud Services & APIs	15,000.00
Internet & Connectivity	4,000.00
Learning Materials	5,000.00
Human Effort Cost	0.00
Total Estimated Budget	24,000.00

Table 2: Budget Plan

7.1 Justification for Each Expense

- **Hardware Costs (0.00 LKR):** As this is a software-centric module, all programming, algorithm development, and initial localized testing will be conducted using the existing personal computers of the research group members. No dedicated physical servers or specialized external hardware are required for this phase.
- **Software Licenses (0.00 LKR):** The core novelty of this component relies on its transparent, modifiable architecture. By strictly utilizing open-source tools—such as Python for core programming, Hugging Face for NLP pipelines, and open-source front-end charting libraries—we completely eliminate the need for expensive proprietary software licenses.
- **Cloud Services & APIs (15,000.00 LKR):** This is the primary source of cost for the project. Generating the foundational synthetic dataset requires interacting with advanced models like Gemini 3 Pro and Deepseek via Knowledge Distillation. Furthermore, running the LLM-driven schema to extract key entities and relationships from user notes requires the processing capacity of cloud-based virtual machines, as standard local hardware is insufficient for high-speed, real-time NLP inference.
- **Internet & Connectivity (4,000.00 LKR):** Continuous, high-bandwidth internet access is required to interface with cloud environments, download heavy public benchmarking datasets like "EdNet" and "ASSISTments", and support real-time collaborative feature testing.

- **Learning Materials (5,000.00 LKR):** To gain the essential specialized domain expertise required for this project, funds are allocated for enrolling in specialized online courses (e.g., on Coursera or edX focusing on AI and Learning Analytics) and purchasing access to paywalled industry research reports or specific IEEE/ACM journal papers.
- **Human Effort Cost (0.00 LKR):** Because this system is being developed as a mandatory IT4010 final-year research project, the labor, coding, and testing hours contributed by the undergraduate researchers are strictly considered academic effort and are not financially billed to the project budget.

8. WORK BREAKDOWN STRUCTURE (WBS)

To ensure the successful delivery of the "AI-Powered Personalized Course Creation and Discovery Platform," the project is modularized into four distinct, yet interconnected, AI components. The Work Breakdown Structure (WBS) is designed to allocate an end-to-end development pipeline to each researcher, ensuring equal technical complexity and a balanced workload.

8.1 Task Breakdown Per Member

- Member 1: Kojithan P.Y (IT22264220) – Interactive Infographic Generator
 - Task 1.1: Design a schema to extract and structure key entities, relationships, and statistics from user notes using an open-source Large Language Model (LLM).
 - Task 1.2: Implement multiple visualization templates (hierarchical, timeline, network) using open-source libraries.
 - Task 1.3: Develop a dynamic interface allowing users to filter data, click elements to drill down into source quotes, and toggle visualization types.
 - Task 1.4: Integrate a rule-based or lightweight AI model to suggest the most suitable chart type.
 - Task 1.5: Enable real-time collaborative features for simultaneous viewing and annotation.
 - Task 1.6: Create an export module for interactive HTML or static image publishing.
- Member 2: Baskaran V (IT22172600) – Interactive AI-Powered Audio Review Component
 - Task 2.1: Implement automatic speech recognition (ASR) to transcribe uploaded audio/podcasts.
 - Task 2.2: Develop an NLP pipeline using an open-source LLM to analyze transcripts for key concepts, structured summaries, and central themes.
 - Task 2.3: Build an interactive web interface with clickable transcript segments for context-specific Q&A.
 - Task 2.4: Integrate an open-source Text-to-Speech (TTS) model for verbal AI answers.
 - Task 2.5: Conduct system evaluation based on transcription accuracy, response latency, and user engagement.
- Member 3: Nishara T (IT22223876) – Adaptive Quiz and Assessment Generation Engine
 - Task 3.1: Develop a Knowledge Tracing Model to infer concept mastery from interaction history.
 - Task 3.2: Implement an NLP pipeline to auto-generate diverse question types (MCQ, fill-in, short answer).
 - Task 3.3: Create an algorithm to dynamically adjust quiz difficulty and focus based on the learner model.

- Task 3.4: Integrate a feedback mechanism offering Socratic hints and scraped web study resources.
- Task 3.5: Evaluate the engine's impact on learning efficacy via pre/post-test scores.
- Member 4: Sarmitha S (IT22637482) – Multilingual Handwritten Notes Extractor
 - Task 4.1: Collect handwritten notes across English, Tamil, and Sinhala.
 - Task 4.2: Preprocess images (denoise, adjust contrast) and prepare labeled training data.
 - Task 4.3: Train or fine-tune an AI model for multilingual handwriting recognition.
 - Task 4.4: Clean, structure, and convert extracted text into study-ready Canvanotes.
 - Task 4.5: Store structured outputs in a searchable digital knowledge base.

8.2 Timeline Alignment: The project spans the 2026 academic year and is divided into four synchronized phases to ensure all components integrate smoothly into the Central Knowledge Orchestrator:

1. Phase 1: Planning & Data Collection (Months 1-2): Literature review, UI/UX design, synthetic data generation, and gathering multilingual note datasets.
2. Phase 2: Core Model Development (Months 3-5): Independent development of LLM extraction schemas, knowledge tracing models, and handwriting recognition algorithms.
3. Phase 3: Integration & UI Implementation (Months 6-7): Connecting the four standalone AI modules to the unified Personalized Learning Interface dashboard.
4. Phase 4: Testing & Pilot Study (Months 8-10): Executing the controlled pilot study with undergraduate volunteers to evaluate performance, followed by ethical clearance reporting and final documentation.

8.3 Balanced Workload Distribution: The workload is equitably distributed by assigning each member a complete, vertically integrated subsystem. Every researcher is responsible for their module's data ingestion (e.g., text, audio, images), underlying AI/ML processing (e.g., NLP extraction, speech recognition, computer vision), user interface integration, and individual performance evaluation. This ensures that all four members engage deeply with core AI domains and software engineering practices without creating technical bottlenecks or unequal dependencies

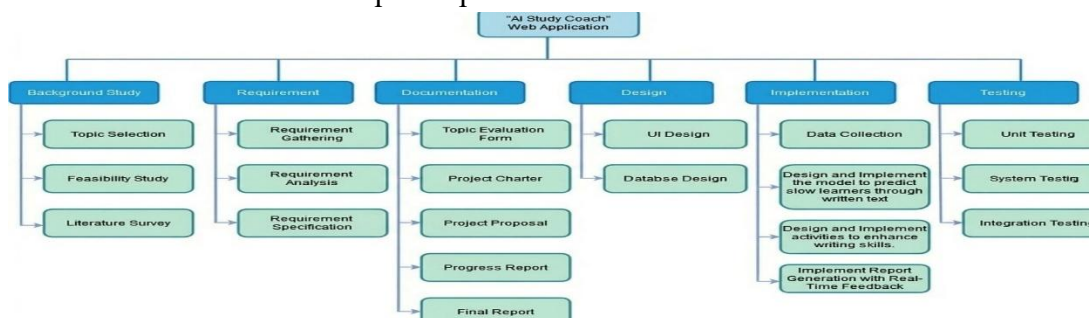


Figure 2:WBS

9. GANTT CHART

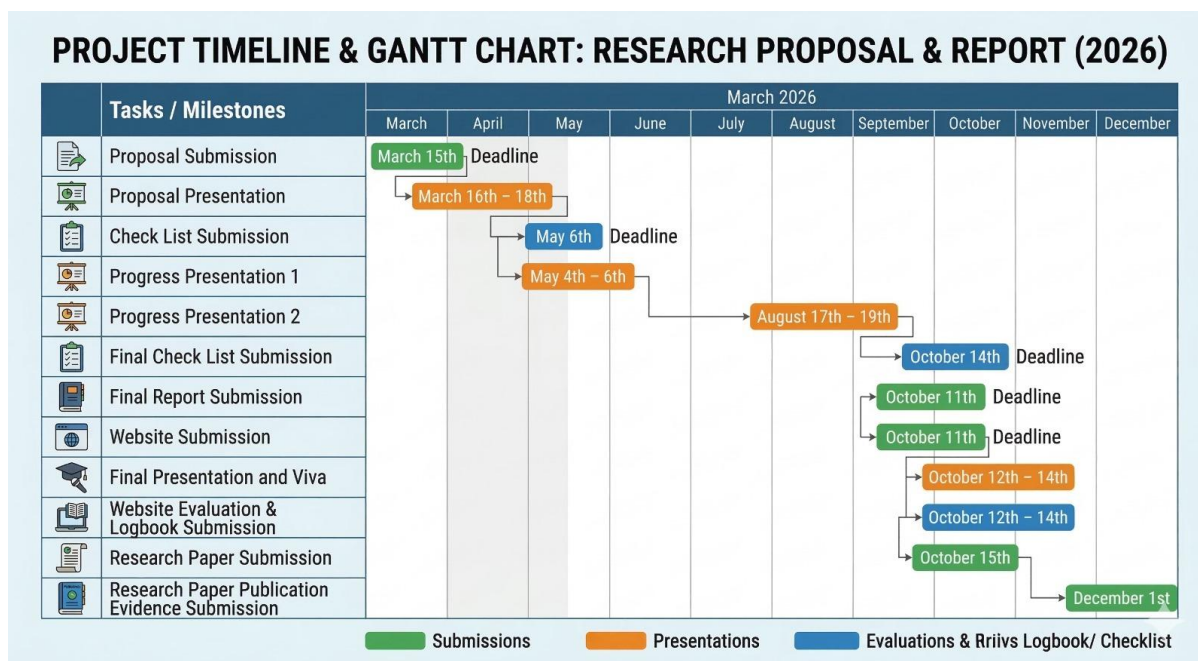


Figure 3: GANTT CHART

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11.APPENDICES

11.1 Survey instruments

1. Online Questionnaires (via Google Forms)

- **What it is:** Digital surveys given to students before and after using your interactive charts.
- **Why use it:** To collect hard numbers and quantitative data (e.g., rating usability on a scale of 1to5).
- **Example:** "Rate how easy it was to switch from the Timeline View to the Network View."

2. Structured Interviews

- **What it is:** A one-on-one conversation using a strict, pre-written list of specific questions.
- **Why use it:** To make sure you get qualitative answers on specific features you built (like the AI chart recommendation).
- **Example:** "Did the AI correctly suggest the best chart type for your notes? Please explain why or why not."

3. Unstructured Interviews

- **What it is:** An open, free-flowing conversation with the user, without a strict script.
- **Why use it:** To discover unexpected insights. Sometimes users interact with a visual chart in ways you didn't plan for. This allows the user to freely share their feelings, frustrations, or "aha!" moments.
- **Example:** "Tell me about your experience exploring the data today. What stood out to you?" (And then you just listen and let the conversation flow naturally).

11.2 Detailed diagrams

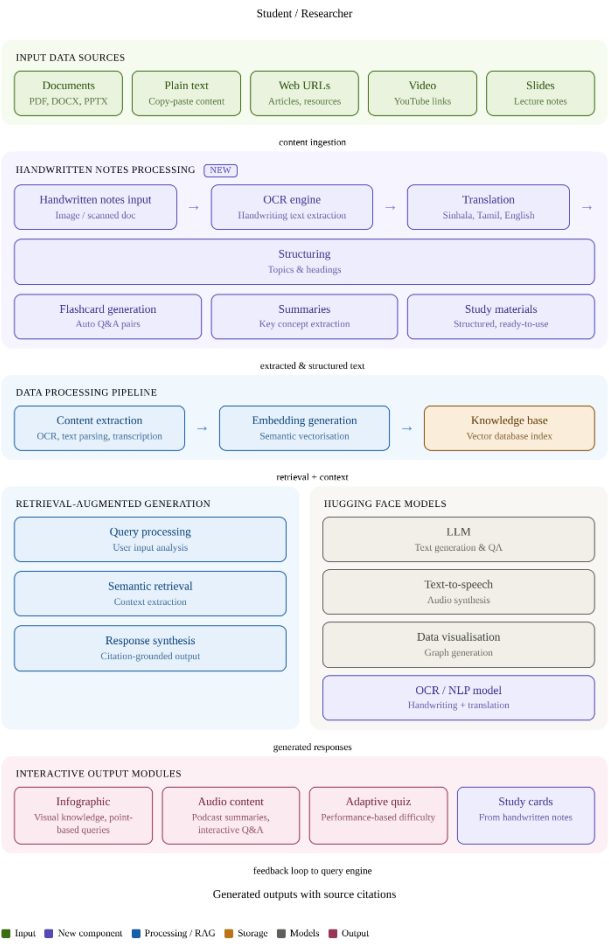


Figure 4: Detailed diagram

11.3 Preliminary UI mockups



Figure 5:Mockup1

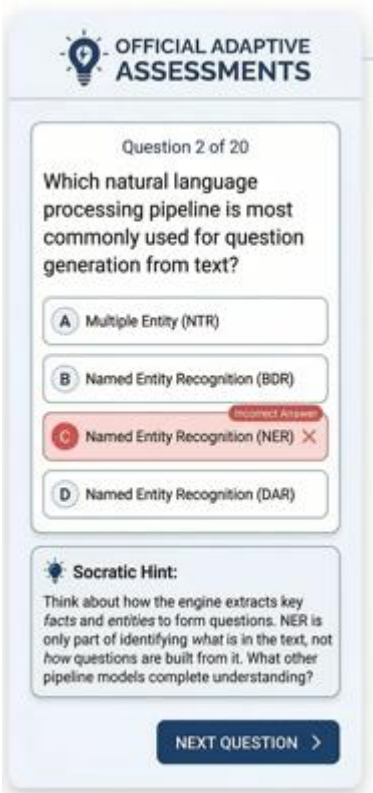


Figure 6:Mockup2



Figure 7:Mockup3

11.4 Risk analysis

For the development of the Adaptive Quiz and Assessment Generation Engine, the following key risks and their respective mitigation strategies have been identified:

1. Technical Risk: NLP Hallucinations and Inaccuracy

- **Risk:** The open-source NLP models used for question generation might occasionally create irrelevant, factually incorrect, or poorly structured questions and Socratic hints from the uploaded student notes.
- **Mitigation:** To address this, strict prompt engineering and response validation layers will be implemented. The models will be rigorously tested and calibrated using generated synthetic data before deployment.

2. Algorithmic Risk: Inaccurate Knowledge Tracing

- **Risk:** The Knowledge Tracing model might fail to accurately evaluate a student's true mastery level, resulting in the system serving questions that are either too easy (causing boredom) or too difficult (causing frustration).
- **Mitigation:** The algorithm will be trained and calibrated using established public educational benchmarking datasets, such as "EdNet" and "ASSISTments," to ensure high accuracy in predicting student performance.

3. Performance Risk: System Latency

- **Risk:** Performing real-time NLP question generation, running the knowledge tracing algorithm, and fetching external web resources (RAG) simultaneously could lead to system delays, disrupting the student's learning flow.
- **Mitigation:** The system architecture will be optimized by minimizing heavy API calls, utilizing efficient cloud services for NLP inference, and maintaining a lightweight database structure (MongoDB) for fast retrieval.

4. Ethical and Privacy Risk: Data Mishandling

- **Risk:** During the controlled pilot study, there is a risk of mismanaging or exposing sensitive student interaction data and performance scores.
- **Mitigation:** The project will strictly adhere to the principle of Data Minimization, logging only essential performance metrics without exposing personal identities. Furthermore, formal ethical clearance will be obtained from the Institutional Review Board (IRB), ensuring informed consent and the right to withdraw for all participants.